

Othman Echi

Professor of Mathematics

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Profiles: [KFUPM Pure](#) | [Google Scholar](#) | [DBLP](#) | [Mathematics Genealogy](#)



Profile

Professor of Mathematics with a research programme spanning commutative algebra, general and spectral topology, combinatorics on words, elementary number theory, digital topology, category theory, and theoretical computer science. Sustained output of journal articles and book chapters; long record of doctoral and habilitation supervision; ongoing editorial, reviewing, and conference service at the international level.

Personal Information

Date and place of birth 20 February 1962, El Hamma Djerid, Tunisia

Nationality Tunisian

Marital status Married

Research Areas

Major fields. Commutative algebra; general topology; combinatorics; elementary number theory; theory of categories.

Research interests. Commutative algebra; combinatorics on words; discrete mathematics and its applications; digital topology; elementary number theory and its applications; foliation theory; general topology; theory of binary relations; theory of categories; theoretical computer science.

Education

Degree	Year	Institution	Country	Specialisation
BSc (Maîtrise)	1986	University of Tunis II	Tunisia	Mathematics
Master (DEA)	1988	University of Tunis II	Tunisia	Algebra
Doctorate of Specialty	1990	University of Tunis	Tunisia	Algebra
Doctorate of State	1993	University of Tunis	Tunisia	Algebra
Habilitation	2001	Université Aix-Marseille III	France	Algebra and topology

Doctoral and Habilitation Theses

- Doctorate of Specialty* — C-rings, E-rings and the altitude formula.
Advisors: Jean-Paul Cahen and Ezzeddine Salhi.
- Doctorate of State* — On chain conditions in the prime spectrum of a commutative ring.

Advisors: Jean-Paul Cahen and Ezzeddine Salhi.

- *Habilitation* — Dimension theory, spectral topology and foliations.

Advisors: Jean-Paul Cahen and Ezzeddine Salhi.

Awards and Scholarships

Year	Distinction
1982	First Prize, graduating with excellence from high school, School of Abul-Kacem Chebbi, Tozeur.
1986	First Prize, graduating with excellence from the École Normale Supérieure, Bizerte.
1986–1988	Tunisian Government Scholarship for graduate studies.

Languages

Language	Reading	Writing	Conversation
Arabic	Excellent	Excellent	Excellent
French	Excellent	Excellent	Excellent
English	Excellent	Excellent	Good

Professional Experience

Position	Period	Institution	Country
Assistant (Lecturer)	1988–1991	Faculty of Sciences of Sfax	Tunisia
Assistant Professor	1991–1994	Faculty of Sciences of Sfax	Tunisia
Associate Professor	1994–2000	Faculty of Sciences of Sfax	Tunisia
Professor	Mar. 2001–present	Faculty of Sciences of Tunis, University of Tunis El Manar	Tunisia
Professor	Sept. 2009–present	King Fahd University of Petroleum & Minerals	Saudi Arabia

Teaching

Undergraduate

Calculus; general topology; differential geometry; Hilbertian analysis; discrete mathematics; linear algebra; abstract algebra; rings and fields; group theory; number theory.

Graduate

- Dimension theory — University of Sfax (1994/1995).
- Commutative algebra — University of Sfax (1995/1996).
- Spectral spaces — University of Sfax (1997/1998).
- Spectral spaces — University of Tunis El Manar (2001/2002).
- Theory of categories and topology — University of Tunis El Manar (2002/2003, 2003/2004).
- Discrete mathematics and applications — University of Tunis El Manar (2005/2006).
- Lattice theory and applications to theoretical computer science — University of Tunis El Manar

(2008/2009).

- MATH 551 Abstract Algebra — KFUPM (Term 092).
- MATH 555 Commutative Algebra — KFUPM (Term 131).
- MATH 521 General Topology I — KFUPM (Terms 132, 251).
- MATH 525 Graph Theory — KFUPM (Term 222).
- MATH 645 Enumerative Combinatorics — KFUPM (Terms 241, 252).

Research Supervision

- Master's theses supervised: **9**.
- Ph.D. and habilitation theses supervised: **13**.
- Mathematics Genealogy Project record: **11** doctoral / habilitation students and **13** descendants.

Doctoral and Habilitation Students (Mathematics Genealogy Project)

Student	Institution	Year	Descendants
Bouacida, Ezzeddine	University of Sfax	1998	—
Ben Nasr, Mabrouk	University of Sfax	1999	1
Cherif, Bahri	University of Tunis	1999	—
Jarboui, Noomen	University of Sfax	2000	1
Belaid, Karim	University of Tunis	2001	1
Gargouri, Riyadh	University of Tunis	2006	—
Lazaar, Sami	University of Tunis	2008	2
Dridi, Lobna	Université Tunis-El Manar	2012	—
Ghanmi, Nejib	Université Tunis-El Manar	2013	—
Ouelld Abdollahi, Mohamed	Université Tunis-El Manar	2016	—
Turki, Tarek	Université Tunis-El Manar	2017	—

Research Projects

Period	Role	Project
2018–2019	Co-investigator	Topological and algebraic theory of convex subrings of ${}^*\mathbb{R}$.
2014	Co-investigator	Unifying Zariski-like topologies for modules, Part II.
2014	Co-investigator	Unifying Zariski-like topologies for modules, Part I.
2010–2011	Principal investigator	A generalisation of Carmichael numbers.
2010–2011	Principal investigator	A categorical study of some typical examples of Alexandroff spaces.

International Conferences and Invited Talks

- Meetings of the Tunisian Mathematical Society — April 1993, March 1994, March 1995, March 1996, March 1997, March 1998, March 2006, March 2007, March 2008.
- First International Conference *Autour des polynômes à valeurs entières* — CIRM, Marseille, France, 1990.
- First International Conference on Commutative Ring Theory — Fez, Morocco, 1992.

- International Conference *Algèbre et applications* — CIRM, Marseille, France, 1993.
- Second International Conference on Commutative Ring Theory — Fez, Morocco, 1995.
- CIMPA International School *Real and Complex Foliations* — Sfax, Tunisia, 11–22 September 1995.
- International Conference *Algèbre et théorie des nombres 1997* — Fez, Morocco, 1997.
- International Conference *Algèbre* — Marseille, France, 1997.
- International Workshop *Commutative Ring Theory III* — Fez, Morocco, 1997.
- Relational Methods in Computer Science (RelMiCS 1997) — Hammamet, Tunisia, 1997.
- Saudi Association for Mathematical Sciences Conference — King Saud University, Riyadh, Saudi Arabia, March 1999.
- International Workshop on Algebra and Applications — KFUPM, Dhahran, Saudi Arabia, 7–9 November 1999.
- Saudi Association for Mathematical Sciences Conference — King Saud University, Riyadh, Saudi Arabia, April 2000.
- 19th Summer Conference on Topology and its Applications — University of Cape Town, South Africa, 5–9 July 2004.
- 6th Pan-African Congress of Mathematicians — Tunis, Tunisia, 1–6 September 2004.
- Fourth International Conference on Applied Mathematics and Computing — Plovdiv, Bulgaria, 12–19 August 2007.
- Fifth International Fez Conference *Commutative Algebra and its Applications* — Fez, Morocco, 23–28 June 2008.
- 24th Summer Conference on Topology and its Applications — Brno, Czech Republic, 14–17 July 2009.
- 2009 International Conference on Topology and its Applications — Hacettepe University, Ankara, Turkey, 6–11 July 2009.
- First International Conference on Mathematics and Statistics (AUS-ICMS'10) — American University of Sharjah, UAE, 18–21 March 2010.
- *Algebra Meets Topology: Advances and Applications* — Conference in honour of Dikran Dikranjan's 60th birthday, Barcelona, Spain, 19–23 July 2010.
- International Conference on Applied Analysis and Algebra — Yildiz Technical University, Istanbul, Turkey, 29 June–2 July 2011.
- International Conference on Applied Analysis and Algebra — Yildiz Technical University, Istanbul, Turkey, 29 June–2 July 2012.
- International Conference on Applied Analysis and Mathematical Modelling (ICAAMM 2017) — Istanbul, Turkey, 3–7 July 2017.
- 33rd Summer Conference on Topology and its Applications — Western Kentucky University, Bowling Green, USA, 17–20 July 2018.
- Second International Spring School on Mathematics: Algebra, Analysis, Topology and Applications — University of Sharjah, UAE, 24–25 March 2019.
- 2019 Interdisciplinary Colloquium in Topology and its Applications — University of Vigo, Spain, 19–22 June 2019.
- International Conference on Emerging Trends in Pure and Applied Mathematics — Kingdom of Bahrain, 21–22 June 2021.
- Conference on Rings and Polynomials — TU Graz, Austria, 19–24 July 2021.
- WORDS'21, 13th International Conference on Words — Rouen, France, 13–17 September 2021.
- 37th Summer Conference on Topology and its Applications — Youngstown State University, USA, 17–21 July 2023.
- 12th International Conference on Applied Analysis and Mathematical Modeling (ICAAMM 2024) — Istanbul, Turkey, 19–23 July 2024.

Academic Leadership and Service

- Elected member, Scientific Council, Faculty of Sciences of Sfax, 1993–1996.
- Elected member, National Jury for Recruitment of Assistants, Ministry of Higher Education, Tunisia, 1995–1999.
- Elected member, National Jury for Recruitment of Assistant Professors, Ministry of Higher Education, Tunisia, 1995–1999 and 2004–2005.
- Vice-President, Tunisian Mathematical Society, 1997–1998.
- Director, Research Group “Algebra”, University of Sfax (E03/C15), 1992–1998.
- Director, Research Unit “Algebra and Topology”, University of Tunis El Manar (03/UR/15-03), 2003–2014.

Editorial and Reviewing Service

- **Editor**, *Arabian Journal of Mathematics*.
- **Reviewer**, zbMATH Open — [rv:00012576](#).
- **Reviewer**, Mathematical Reviews / AMS MathSciNet.
- **Referee** for journals and series including: *AJSE Mathematics*; *African Diaspora Journal of Mathematics*; *Allahabad Mathematical Society*; *Bulletin of the Malaysian Mathematical Sciences Society*; *Communications in Algebra*; *International Journal of Mathematics and Mathematical Sciences*; *Lecture Notes in Pure and Applied Mathematics*; *Matematicki Vesnik*; *Monatshefte für Mathematik*; *Topology and its Applications*; *Beiträge zur Algebra und Geometrie / Contributions to Algebra and Geometry*; *Journal of Number Theory*; *Journal of Mathematical Analysis*; *Turkish Journal of Mathematics*; *Quaestiones Mathematicae*; *International Journal of Mathematical Education in Science and Technology*; *AIMS Mathematics*; *Filomat*; *ACM Transactions on Knowledge Discovery from Data*.
- **External evaluator** for institutions including: University of Sana’a, University of Tunis El Manar, University of Sfax, University of Dammam, Umm Al-Qura University, King Saud University, Taibah University, and University of Monastir.

Publications and Scholarly Contributions

MSC areas represented. General topology (54-XX); commutative algebra (13-XX); order, lattices and ordered algebraic structures (06-XX); category theory and homological algebra (18-XX); computer science (68-XX); number theory (11-XX); combinatorics (05-XX); manifolds and cell complexes (57-XX); sequences and series (40-XX); logic and foundations (03-XX); field theory and polynomials (12-XX); group theory (20-XX); real functions (26-XX).

Author entries are listed in chronological order. The author’s name is rendered in bold throughout.

- [1] **Echi, Othman**. C-anneaux, E-anneaux et formule de la dimension. *Portugaliae Mathematica* **50** (1993), no. 3, 277–295.
- [2] **Echi, Othman**. Transfert de la notion de C-anneaux aux produits fibrés. *Portugaliae Mathematica* **51** (1994), no. 1, 13–23.
- [3] **Echi, Othman**. The altitude formula. In *Commutative Ring Theory: Proceedings of the Fes International Conference*, Lecture Notes in Pure and Applied Mathematics **153**, Marcel Dekker, 1994, pp. 99–104. [DOI: 10.1201/9781003421917-10](#).
- [4] Ayache, A.; Cahen, P.-J.; **Echi, Othman**. Intersection de produits fibrés et formule de la dimension. *Communications in Algebra* **22** (1994), no. 9, 3495–3509.
- [5] **Echi, Othman**. Sur les hauteurs valuatives. *Bollettino della Unione Matematica Italiana B (7)* **9** (1995), no. 2, 281–297.
- [6] Ayache, A.; Cahen, P.-J.; **Echi, Othman**. Valuation heights and infinite Nagata rings. *Communications in Algebra* **23** (1995), no. 5, 1913–1926.

- [7] Ayache, A.; Cahen, P.-J.; **Echi, Othman**. Anneaux quasi-préfériens et P -anneaux. *Bollettino della Unione Matematica Italiana B (7)* **10** (1996), no. 1, 1–24.
- [8] Bouacida, E.; **Echi, Othman**; Salhi, E. Topologies associées à une relation binaire et relation binaire spectrale. *Bollettino della Unione Matematica Italiana B (7)* **10** (1996), no. 2, 417–439.
- [9] Belaïd, K.; Cherif, B.; **Echi, Othman**. On spectral binary relations. In *Commutative Ring Theory*, Lecture Notes in Pure and Applied Mathematics **185**, Marcel Dekker, 1997, pp. 79–88.
- [10] Bouacida, E.; **Echi, Othman**; Salhi, E. Nonfinite heights. In *Commutative Ring Theory*, Lecture Notes in Pure and Applied Mathematics **185**, Marcel Dekker, 1997, pp. 113–130.
- [11] Bouacida, E.; **Echi, Othman**; Salhi, E. Foliations, spectral topology, and special morphisms. In *Advances in Commutative Ring Theory*, Lecture Notes in Pure and Applied Mathematics **205**, Marcel Dekker, 1999, pp. 111–132.
- [12] Ben Nasr, M.; **Echi, Othman**; Izelgue, L.; Jarboui, N. Pairs of domains where all intermediate domains are Jaffard. *Journal of Pure and Applied Algebra* **145** (2000), no. 1, 1–18.
- [13] Ameziane, S.; **Echi, Othman**; Yengui, I. Direct systems of localizations of polynomial rings. *Acta Scientiarum Mathematicarum (Szeged)* **66** (2000), no. 3–4, 465–476.
- [14] Bouacida, E.; **Echi, Othman**; Salhi, E. Feuilletages et topologie spectrale. *Journal of the Mathematical Society of Japan* **52** (2000), no. 2, 447–464.
- [15] **Echi, Othman**. A topological characterization of the Goldman prime spectrum of a commutative ring. *Communications in Algebra* **28** (2000), no. 5, 2329–2337.
- [16] Belaid, K.; Cherif, B.; **Echi, Othman**. Quasi-spectral binary relations and ordered disjoint unions. *Journal of Mathematical Sciences (Calcutta)* **11** (2000), no. 2, 139–157.
- [17] Ayache, A.; Ben Nasr, M.; **Echi, Othman**; Jarboui, N. Universally catenarian and going-down pairs of rings. *Mathematische Zeitschrift* **238** (2001), no. 4, 695–731.
- [18] **Echi, Othman**; Jarboui, N. On residually integrally closed domains. *Demonstratio Mathematica* **36** (2003), no. 3, 543–550.
- [19] **Echi, Othman**. Quasi-homeomorphisms, Goldspectral spaces and Jacspectral spaces. *Bollettino della Unione Matematica Italiana B (8)* **6** (2003), no. 2, 489–507.
- [20] **Echi, Othman**. Topological characterizations of some subspaces of a spectral space. *Questions and Answers in General Topology* **21** (2003), no. 2, 109–123.
- [21] Bouacida, E.; **Echi, Othman**; Picavet, G.; Salhi, E. An extension theorem for sober spaces and the Goldman topology. *International Journal of Mathematics and Mathematical Sciences* **2003**, no. 51, 3217–3239.
- [22] Belaid, K.; **Echi, Othman**; Gargouri, R. A-spectral spaces. *Topology and its Applications* **138** (2004), no. 1–3, 315–322. DOI: [10.1016/j.topol.2003.08.009](https://doi.org/10.1016/j.topol.2003.08.009).
- [23] Belaid, K.; **Echi, Othman**. On a conjecture about spectral sets. *Topology and its Applications* **139** (2004), no. 1–3, 1–15.
- [24] **Echi, Othman**; Gargouri, R. An up-spectral space need not be A-spectral. *New York Journal of Mathematics* **10** (2004), 271–277.
- [25] Belaid, K.; **Echi, Othman**; Lazaar, S. $T(\alpha, \beta)$ -spaces and the Wallman compactification. *International Journal of Mathematics and Mathematical Sciences* **2004**, no. 68, 3717–3735. DOI: [10.1155/S0161171204404050](https://doi.org/10.1155/S0161171204404050).
- [26] **Echi, Othman**; Ito, M. Up-spectral spaces and A-spectral spaces. *Questions and Answers in General Topology* **23** (2005), no. 1, 15–26.
- [27] Belaid, K.; **Echi, Othman**; Gargouri, R. Two classes of locally compact sober spaces. *International Journal of Mathematics and Mathematical Sciences* **2005**, no. 15, 2421–2427. DOI: [10.1155/IJMMS.2005.2421](https://doi.org/10.1155/IJMMS.2005.2421).
- [28] Ayache, A.; **Echi, Othman**. The envelope of a subcategory in topology and group theory. *International Journal of Mathematics and Mathematical Sciences* **2005**, no. 21, 3387–3404. DOI: [10.1155/IJMMS.2005.3387](https://doi.org/10.1155/IJMMS.2005.3387).
- [29] Ayache, A.; Dobbs, D. E.; **Echi, Othman**. Reflection of some quasi-local domains. *Journal of Algebra and its Applications* **5** (2006), no. 2, 201–213.
- [30] Ayache, A.; **Echi, Othman**. Valuation and pseudo-valuation subrings of an integral domain. *Communications in Algebra* **34** (2006), no. 7, 2467–2483. DOI: [10.1080/00927870600650515](https://doi.org/10.1080/00927870600650515).
- [31] Ayache, A.; Dobbs, D. E.; **Echi, Othman**. Universal mapping properties of some pseudovaluation domains and related quasilocal domains. *International Journal of Mathematics and Mathematical Sciences* **2006**,

- Art. ID 72589, 12 pp.
- [32] **Echi, Othman.** On T_D -spaces. *Missouri Journal of Mathematical Sciences* **18** (2006), no. 1, 54–60. DOI: [10.35834/2006/1801054](https://doi.org/10.35834/2006/1801054).
 - [33] **Echi, Othman.** Binomial coefficients and Nasir al-Din al-Tusi. *Scientific Research and Essay* **1** (2006), no. 2, 28–32.
 - [34] **Echi, Othman.** Networks of morphisms. *Missouri Journal of Mathematical Sciences* **19** (2007), no. 2, 93–105.
 - [35] Ayache, A.; Dobbs, D. E.; **Echi, Othman.** On maximal non-ACCP subrings. *Journal of Algebra and its Applications* **6** (2007), no. 5, 873–894.
 - [36] **Echi, Othman;** Marzougui, H.; Salhi, E. Problems from the Bizerte-Sfax-Tunis Seminar. In *Open Problems in Topology II*, E. Pearl (ed.), Elsevier, 2007, pp. 669–674.
 - [37] **Echi, Othman.** Williams numbers. *Mathematical Reports of the Academy of Sciences of the Royal Society of Canada* **29** (2007), no. 2, 41–47.
 - [38] **Echi, Othman;** Lazaar, S. Sober spaces and sober sets. *Missouri Journal of Mathematical Sciences* **20** (2008), no. 1, 60–72. DOI: [10.35834/mjms/1316032836](https://doi.org/10.35834/mjms/1316032836).
 - [39] **Echi, Othman;** Gargouri, R.; Lazaar, S. On the Hochster dual of a topological space. *Topology Proceedings* **32** (2008), 153–166.
 - [40] Adams, M. E.; Belaid, K.; Dridi, L.; **Echi, Othman.** Submaximal spaces and spectral spaces. *Mathematical Proceedings of the Royal Irish Academy* **108A** (2008), 137–147.
 - [41] **Echi, Othman;** Naimi, M. Primitive words and spectral spaces. *New York Journal of Mathematics* **14** (2008), 719–731.
 - [42] **Echi, Othman;** Lazaar, S. Universal spaces, Tychonoff and spectral spaces. *Mathematical Proceedings of the Royal Irish Academy* **109A** (2009), no. 1, 35–48. DOI: [10.3318/PRIA.2008.109.1.35](https://doi.org/10.3318/PRIA.2008.109.1.35).
 - [43] **Echi, Othman;** Lazaar, S. Reflective subcategories, Tychonoff spaces and spectral spaces. *Topology Proceedings* **34** (2009), 307–319.
 - [44] **Echi, Othman.** The cardinality of a sphere relative to an edit distance. *Applied Mathematical Sciences* **3** (2009), no. 17–20, 813–824.
 - [45] **Echi, Othman;** Lazaar, S. Quasihomomorphisms and lattice equivalent topological spaces. *Applied General Topology* **10** (2009), no. 2, 227–237. DOI: [10.4995/agt.2009.1736](https://doi.org/10.4995/agt.2009.1736).
 - [46] Bouallègue, K.; **Echi, Othman;** Pinch, R. Korselt numbers and sets. *International Journal of Number Theory* **6** (2010), no. 2, 257–269.
 - [47] Ayache, A.; **Echi, Othman;** Naimi, M. On Kronecker polynomials. *Rocky Mountain Journal of Mathematics* **41** (2011), no. 3, 707–725. DOI: [10.1216/RMJ-2011-41-3-707](https://doi.org/10.1216/RMJ-2011-41-3-707).
 - [48] **Echi, Othman;** Oued Abdallahi, M. On the spectralification of a hemispectral space. *Journal of Algebra and its Applications* **10** (2011), no. 4, 687–699. DOI: [10.1142/S0219498811004847](https://doi.org/10.1142/S0219498811004847).
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 - [50] **Echi, Othman;** Ghanmi, N. The Korselt set of pq . *International Journal of Number Theory* **8** (2012), no. 2, 299–309.
 - [51] **Echi, Othman.** The categories of flows of Set and Top. *Topology and its Applications* **159** (2012), no. 9, 2357–2366. DOI: [10.1016/j.topol.2011.11.059](https://doi.org/10.1016/j.topol.2011.11.059).
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 - [53] Al-Rasasi, I.; **Echi, Othman;** Ghanmi, N. On the Korselt set of a squarefree composite number. *Mathematical Reports of the Academy of Sciences of the Royal Society of Canada* **35** (2013), 1–15.
 - [54] Bakhtary, P.; **Echi, Othman.** On minimal Hamming compatible distances. *RAIRO — Theoretical Informatics and Applications* **48** (2014), no. 5, 495–503. DOI: [10.1051/ita/2014022](https://doi.org/10.1051/ita/2014022).
 - [55] **Echi, Othman;** Lazaar, S.; Oued Abdallahi, M. On some orthogonal factorization systems. *Journal of Algebra and its Applications* **14** (2015), no. 8, 1550120, 18 pp.
 - [56] **Echi, Othman.** Non-primitive words of the form pq^m . *RAIRO — Theoretical Informatics and Applications* **51** (2017), no. 3, 141–166. DOI: [10.1051/ita/2017012](https://doi.org/10.1051/ita/2017012).

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- [58] **Echi, Othman**; Khalfallah, A. Order theoretic and topological characterizations of the divided spectrum of a ring. *Bulletin of the Belgian Mathematical Society — Simon Stevin* **26** (2019), no. 3, 453–467.
- [59] **Echi, Othman**; Turki, T. On spectral primal spaces. *Journal of Algebra and its Applications* **18** (2019), no. 2, 1950030, 10 pp.
- [60] Ayache, A.; **Echi, Othman**; Khalfallah, A. Subrings of a power set and clopen topologies. *Journal of Algebra and its Applications* **20** (2021), no. 12, Paper No. 2150224, 13 pp.
- [61] **Echi, Othman**. On the product of primal spaces. *Quaestiones Mathematicae* **45** (2022), no. 1, 1–10. DOI: [10.2989/16073606.2020.1836062](https://doi.org/10.2989/16073606.2020.1836062).
- [62] **Echi, Othman**; Khalfallah, A.; Kroumi, D. On primitive words with non-primitive product. *RAIRO — Theoretical Informatics and Applications* **56** (2022), Article 4, 11 pp. DOI: [10.1051/ita/2022004](https://doi.org/10.1051/ita/2022004).
- [63] **Echi, Othman**. On a topology defined by primitive words. *Missouri Journal of Mathematical Sciences* **34** (2022), no. 2, 221–229. DOI: [10.35834/2022/3402221](https://doi.org/10.35834/2022/3402221).
- [64] **Echi, Othman**; Khalfallah, A.; Kroumi, D. Estimating the remainder of an alternating p -series using hypergeometric functions. *Journal of Mathematical Inequalities* **17** (2023), no. 2, 569–580. DOI: [10.7153/jmi-2023-17-36](https://doi.org/10.7153/jmi-2023-17-36).
- [65] **Echi, Othman**; Khalfallah, A.; Kroumi, D. Sharp estimate of the remainder of some alternating series. *Mathematical Inequalities & Applications* **26** (2023), no. 1, 83–91.
- [66] **Echi, Othman**. Quasihomeomorphisms and Skula spaces. *Journal of Algebra and its Applications* **22** (2023), no. 1, Article ID 2350020, 13 pp. DOI: [10.1142/S0219498823500202](https://doi.org/10.1142/S0219498823500202).
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- [68] **Echi, Othman**. The primitive deficiency of two primitive strings. *Acta Informatica* **62** (2025), no. 3, Article 26, 19 pp. DOI: [10.1007/s00236-025-00494-y](https://doi.org/10.1007/s00236-025-00494-y).
- [69] **Echi, Othman**. The group of autohomeomorphisms of some digital topologies on the integers. *Filomat* **39** (2025), no. 9, 2985–3001. DOI: [10.2298/FIL2509985E](https://doi.org/10.2298/FIL2509985E).